PyMC Labs

MARKETING CHANNEL CONTRIBUTION TO REVENUE

Search drove the revenue

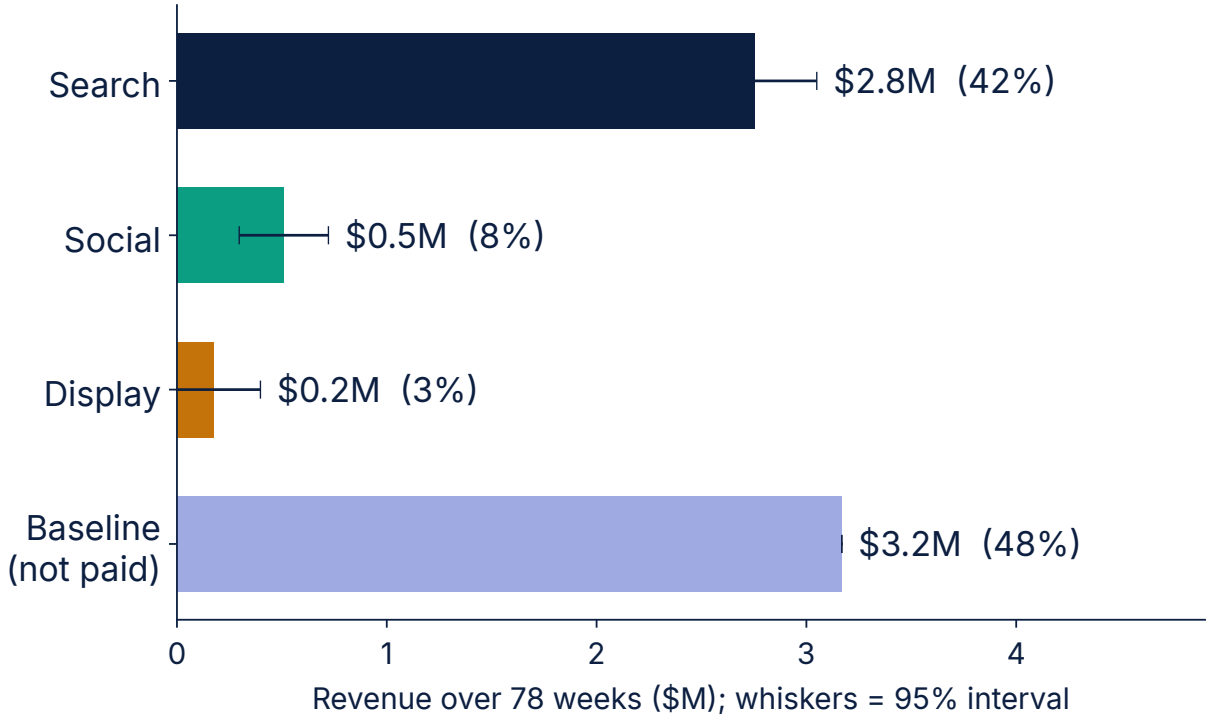
About $3 back for every $1 spent. Social roughly breaks even. Display can’t be told apart from zero.

$3.16
revenue per $1 on search (95%: $2.82 – $3.49)

$1.06
revenue per $1 on social (95%: $0.62 – $1.51)

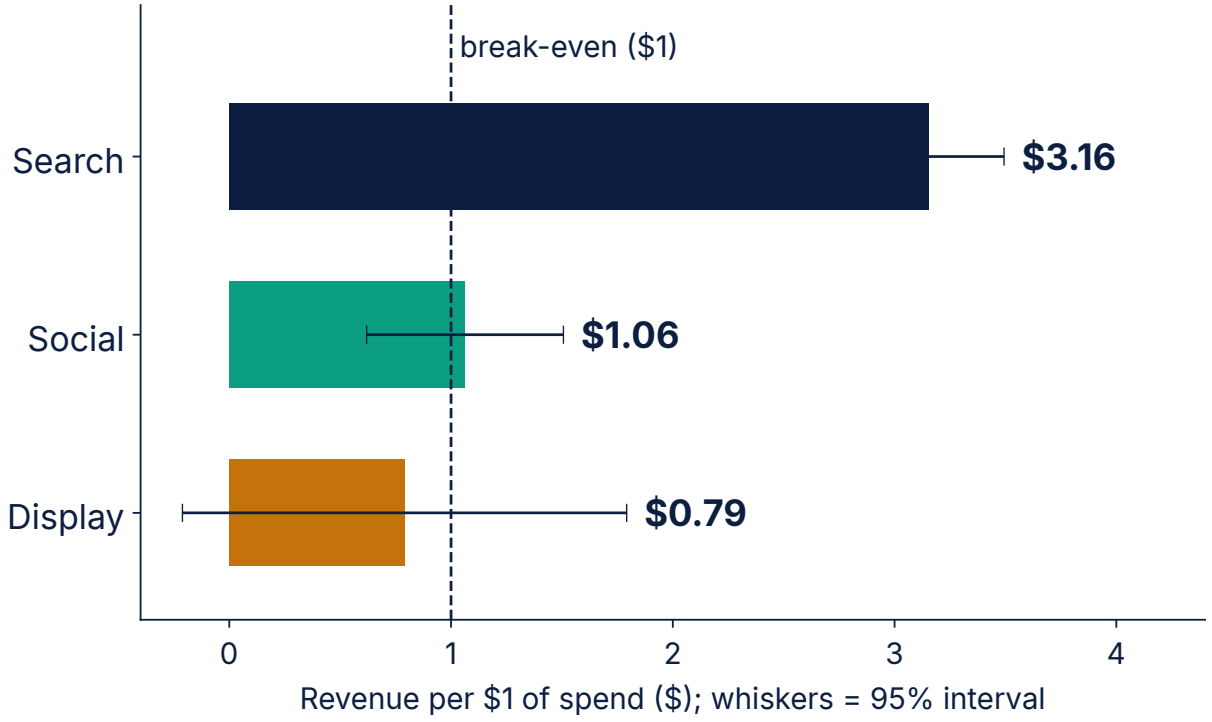
$0.79
revenue per $1 on display (95%: −$0.21 – $1.79)

Search drove 42% of revenue; nearly half has nothing to do with paid spend



- **Search:** $2.8M of $6.6M total, from 55% of paid spend.
- **Social:** $0.5M (8%), from 30% of paid spend.
- **Display:** $0.2M (3%), from 14% of paid spend. The interval spans zero.
- **Baseline:** $3.2M (48%) isn't tied to paid spend.

Search returns about 3× its cost; social breaks even and display is unproven



- **Search** is the only channel whose whole interval sits well above $1.
- **Social** is centred near break-even ($1.06); its interval runs from $0.62 to $1.51.
- **Display** is the least certain: from a loss to nearly 2× return are all plausible.

Channel spends vary independently week to week ($|r| < 0.1$), which is why the three can be separated.

What the data supports, and what it doesn’t

SOLID

Search is the main paid driver, and its estimate is tight.

The linear model explains 83% of weekly revenue variation. Adding diminishing returns didn’t improve the fit.

UNCERTAIN

Display’s true return: the interval runs from −$0.21 to $1.79.

Social is near break-even, so a small shift in the estimate changes the call.

NOT TESTED

These are associations, not a controlled experiment.

No lag or seasonality terms. Search is the largest spend, so saturation matters before scaling it further.

Data checks: 78 consecutive weekly rows, no missing weeks, no duplicates.

Keep funding search, test it, and hold the rest

1. **Keep search funded and run a holdout or geo test on it.** This confirms the $3.16 return and shows where returns start to flatten before any increase.
2. **Hold social at current levels.** It roughly pays for itself; look at creative and targeting, not budget.
3. **Don’t expand display yet.** Run a short on/off test to learn whether it returns anything.
4. **Extend the model with lag and seasonality,** then re-read the channel returns.

Recommendations are PyMC Labs’ proposals from this analysis; they are not outputs of the model.